



COURSE INFORMATION

Course Number and Title: BUAD 5132: Big Data & Cloud Computing Technologies

Credits: 4

Description

This course is designed to provide students with a deep understanding of big data and cloud computing. The data storage and retrieval techniques that have served the information processing industry for decades have proven inadequate in the face of the huge collections of data. Businesses are requiring a new set of technologies that are specifically designed to deal with these huge data sets. This course will focus on applications of big data and cloud computing techniques that will be used to process large-scale data sets.

Course Outcomes

By the end of the course, students will be able to:

- Use MapReduce to process and analyze Big Data
- Use Spark to process and analyze Big Data
- Implement Big Data analytics for various business applications
- Examine cloud computing services
- Analyze Big Data using cloud based services

COURSE DETAILS

Text and Resource List

Required Books

- Asllani, A. (2020). *Big Data Technologies for Business*. Prospect Press. ISBN: 9781943153756

Additional Readings

Articles/materials pertinent to the weeks' topics will be posted for students on Canvas. As the course progresses, timely articles might be added as well.



Course Structure

The course will follow this general pattern:

- Readings, lectures, and quizzes are to be completed early in the module.
- Practice exercises should be completed before starting to work on the programming assignments.
- You should start on your coding/programming assignments no later than midway through each module.
- Review, refine, and submit your coding/programming assignments at the end of each module.
- You will have two group assignments.
- There will be discussion topics in some modules. Students must post both an initial contribution by Thursday midnight and respond to other students' posts by Sunday midnight, unless otherwise posted for that module.

Outline of Course

This course will be broken up into 7-week modules. The following outline presents the topics to be covered in each module.

Module 1: Introduction to Big Data and MapReduce

Module 2: MapReduce Applications

Module 3: Spark

Module 4: Spark Applications

Module 5: SparkSQL

Module 6: Cloud Computing

Module 7: Big Data Applications on Cloud

Grading Methodology

Performance in this course will be assessed through completion of the following activities. The [Honor Code Categories](#) noted in this table describe the acceptable forms of interaction that are permitted in the process of completing each assignment.



Assignment Name	Points	Percentage	Team/ Individual	Honor Code Category
Programming Assignments	655	65.5%	Individual	A
Group Assignments	200	20%	Team	B
Quizzes	75	7.5%	Individual	A
Discussion Board	50	5%	Individual	A
Peer Evaluation for Group Assignments	20	2%	Individual	A
Total	1000	100%		

Programming Assignments

In each module, except in Module 6, you will complete a programming assignment. Please carefully review the instructions provided for those assignments. It is important that you spend time working through the practice exercises before attempting the assignments.

Group Assignments

You will have two group assignments throughout the course.

Quizzes

There will be three quizzes. Quizzes will be based on the readings and lecture materials in the module.

Discussions

Class discussions will focus on applying the concepts of the module to real-world situations. You will be expected to make initial posts and respond to your peers. You will be graded on the depth of your posts rather than frequency.

Peer Evaluations

You will fill out a brief group evaluation survey to provide feedback on how well your group worked together.

Please note: Your grade on group assignments may be adjusted based on your peers' feedback.



Grading Scale

Grades are determined by cumulative average point total as follows:

Letter Grade	Range
A	100% to 95%
A-	< 95% to 90%
B+	< 90% to 87%
B	< 87% to 83%
B-	< 83% to 79%
C+	< 79% to 76%
C	< 76% to 73%
C-	< 73% to 70%
F	< 70% to 0%

Late Assignment Policy

It is very important that work be turned in on time or students will find it very difficult to catch up. All work in the course will be due by 11:59 p.m. ET on the date noted in the course calendar, unless noted otherwise.

Any assignment submitted after the due date will have **20%** of the possible points deducted for each day that it is late. Assignments submitted more than **2** days late will receive a grade of 0%. Students should reach out to their instructor immediately to discuss any concerns. Communication with the instructor and teammates (for group assignments) before deadlines are missed is expected.

Submitting Assignments

Completed assignments should be submitted to the appropriate assignment dropbox in Canvas, unless directed otherwise. For group assignments submitted to the Canvas, only one member of the team needs to submit.



Students may be allowed to submit assignments multiple times prior to the deadline; however, the last submission will be graded. Once an assignment is graded, then resubmissions will not be allowed. Students are expected to verify that the correct file was uploaded. Failure to do so will result in a grade of 0%. Correct files submitted after the due date will receive a point deduction according to the Late Assignment Policy.

Citation Expectations

All research work submitted should be properly cited using APA standards. For more information and resources to assist with proper citation style, please refer to the McLeod Business Library at the Mason School [Research Paper Writing and Citing FAQ resources](#).

PROGRAM POLICIES

Academic Policies

An updated [Graduate Student Catalog](#) is released in the Fall of every academic year by the University Registrar's Office in conjunction with the Raymond A. Mason School of Business. Please be sure to select the current academic year graduate catalog from the drop-down menu, and then select the Raymond A. Mason School of Business from the left menu to access and review the school's academic policies and other relevant information.

Because you are held accountable to the academic standards and policies as set forth, you should familiarize yourself with the graduate catalog every year.

Faculty Communication and Feedback

At the beginning of each course, make sure that you understand the instructor's preferred mode of communication and any specific communication protocol. One of the best ways to be effective as a graduate student is to understand the instructor's expectations and operate within those boundaries. You should give the instructor 24 hours to get back to you on any communication and one week for grading turnaround time. If you have concerns about communication or feedback, you should always go to the professor first. You should explain your concern as clearly as possible without judgment or emotion. Effective communication is an important graduate-level skill, and every interaction in your master's program is an opportunity to develop this skill.

Online Learning Student Expectations



All students in this course are expected to have completed the Online Student Orientation course. Through this orientation, you will learn how to navigate the course and use the various tools that you will need in order to participate effectively and submit assignments. You are also expected to have all the equipment and software needed to be successful in the course.

All students are expected to contribute to their own learning as active and well-prepared participants. Activities in each weekly module generally do not require your “electronic presence” at any particular time or day, so there is no class to “miss.” You will have tasks to complete by the end of each week and, possibly, some mid-week deliverables, which you can schedule at your convenience. You should plan to spend the same amount time working on this course as you would if you were taking the course in a face-to-face environment.

Technology Support

Technical support is available using the “Help” link at left in Canvas. Your instructor will always take technical problems into account if the situation warrants it.

Logging in to the Canvas Learning Management System

The learning activities for each week are carefully sequenced so that you can accomplish a reasonable amount of work throughout the week. You should log on to the course website regularly to work through course materials and participate in course discussions.

Technology Requirements

The MSBA program will require the use of many analytics software packages, with R and Python used in multiple courses. Instructions for installing R and Python are provided below. Instructions for other software packages will be included in specific courses that require those software.

The use of a laptop with the Microsoft Windows operating system is mandatory for graduate students at the Raymond A. Mason School of Business at William & Mary. To eliminate possible security and software conflicts, it is highly recommended that your Windows environment be used for Mason coursework only, and not for personal or business purposes. The MSBA faculty have selected the following [laptop available for purchase at this link](#) that meets the computer requirements for the program. You may find it beneficial to have a second monitor to utilize with your laptop.

Posting Responses

Interaction between students is an important part of this course and requires prompt postings and responses. In an attempt to be efficient with our time and considerate of everyone’s schedules—beyond the requirements of this course—we will operate under a consistent time structure for posting assignments and responses to online discussions.



Teamwork

As you advance in your careers, it becomes increasingly important to develop the skill of influencing others and working collaboratively. Similarly, in most workplaces, you will need to collaborate or negotiate with others in order to accomplish their goals.

Teamwork in many courses offers an opportunity for you to learn virtual team leadership and team theory. It is important that you treat your team members with the same respect that you would treat coworkers. Team members will become part of your professional network, so if you work effectively with them, they may become a lifetime resource.

Remember that everyone's style of communication is different, and that it is important to listen carefully and adjust your style when necessary to communicate effectively. You should be proactive when you feel there might be a problem in the group. The situation will not improve if you do not address it.

Academic Integrity and the Honor Code

The origin of our Honor Code at William & Mary dates back to 1736 when the college first demonstrated its highest commitment to the character of its students. The Honor Code has evolved along with the college and has gone from being a "gentlemen's code" into a codified student honor system promoting individual responsibility on the part of all who are members of the William & Mary community relating to a student's honor. The same high commitment to student character exists today, and the Honor Code is administered by the student body to help build a community of trust. Upon matriculation at William & Mary, all students are asked to sign the Honor Code, expressing their commitment to not lie, cheat, or steal.

"As a member of the William and Mary community, I pledge on my honor not to lie, cheat, or steal, either in my academic or personal life. I understand that such acts violate the Honor Code and undermine the community of trust, of which we are all stewards."

Upon graduation, the Honor Code becomes the brand with which all William & Mary alumni will be judged.

Honor and integrity are fundamental assets to the Mason School of Business community, and therefore they must be cherished and protected. All too often today, headlines read of business leaders participating in dishonorable and unethical behavior. Our conduct during our time as graduate students and beyond reflects our character, the character of the Mason School of Business, and our college. Therefore, we encourage students to be actively committed to safeguarding the Honor Code's "community of trust of which we are stewards."

Plagiarism

The following identifies the university policy on plagiarism:



Plagiarism: the presentation, with intent to deceive, or with disregard for proper scholarly procedures of a significant scope, of any information, ideas or phrasing of another as if they were one's own without giving appropriate credit to the original source.

- a) One commits plagiarism when one includes the words of another without quotation or when one includes the substantive work of another without properly crediting the source with footnotes, quotation marks, or other appropriate citation.
- b) A student's intent may be inferred based on the extent and context of the improperly cited material and whether the student has provided false citation or has manipulated the original text such that a reasonable person may conclude the student did so in order to avoid detection.
- c) Disregard for proper scholarly procedure that is minimal in scope may be addressed solely as an academic matter, and the instructor may determine whether an academic penalty should be applied without pursuing resolution under the Honor Code. But any intentional acts of plagiarism or disregard for scholarly procedure of a significant scope should be treated as a violation of the Honor Code and addressed under either Sec. VIII or Sec. IX of the Student Handbook.

Academic Integrity: Categories of Conduct

Academic integrity is an integral component of the William & Mary learning experience, and any breach of this integrity is very serious and not in keeping with the overall intellectual and ethical foundations of our University. Students are expected to adhere to the William & Mary Honor Code and to the general principles of academic honesty. These principles include and incorporate the concept of respect for the intellectual property of others, the expectation that assignments will be submitted according to guidelines specified by the instructor, and that plagiarism of any type is unacceptable.

To clearly delineate acceptable from unacceptable conduct for completing assignments, which are varied in nature due to the varied intended learning outcomes, the Raymond A. Mason School of Business requires that each assignment be identified with the categories below. If there are questions regarding acceptable conduct for a particular assignment, then it is essential that you ask your professor for clarification.

- **Category A** –This is an individual assignment. You may not receive help from anyone on this assignment. It must be 100% your own work. All questions concerning this assignment should be addressed to your professor. It is an honor code offense to give or receive assistance on this assignment.



- **Category B** – This is a group assignment. Your group may not receive help from anyone outside your group. While your group may choose to delegate the work among the group members, everyone in the group is expected to be prepared to discuss the entire assignment in class. All questions concerning this assignment should be addressed to your professor. It is an honor code offense to give help to other groups and individuals or receive assistance from other groups and individuals.
- **Category C** – This is an individual assignment. You may work with others or receive help from a tutor on this assignment. You must, however, turn in your own paper. You may not divide the work with others or copy another student's paper; it is an honor code offense to do so.
- **Category D** – This is a group assignment. You may share information or discuss general concepts and approaches to the assignment with other groups. You may receive help from a tutor. Everyone in the group should be prepared to discuss the entire assignment in class. Each group must turn in their own work. You may not copy another group's work; it is an honor code offense to do so.
- **Category E** – This is a timed assignment. You are given a specific length of time within which the work must be completed. It is an honor code offense to violate this time restriction unless you have received permission from your professor.

Academic Integrity: Use Of Artificial Intelligence (AI)

We recognize the importance and benefits of using Artificial Intelligence (AI) tools, which is why our position is to embrace and integrate them into certain course activities with full transparency about their use. The information and guidelines in this policy ensure that AI is effectively and ethically used in this course.

The following categories were developed within the Mason School of Business to provide a degree of consistency across courses and programs.

- **Category 1 - No AI Assistance:** You may not receive help from any AI tools including editing applications (e.g., Grammarly), generative AI (e.g., ChatGPT), or AI tools integrated within software platforms (e.g., Microsoft Office). All work must be 100% your own. It is an honor code offense to use AI tools for assistance on this assignment in any way.
- **Category 2 - AI for Learning and Practice:** You may use AI tools for learning and practice. This means you can use AI to learn concepts, practice problem-solving, and enhance your understanding of the course material. However, any work submitted for grading must be 100% your own and not the result of AI assistance (i.e., the final work produced must be solely your own). It is an honor code offense to submit work for grading that was produced with the aid of AI.



- **Category 3 - AI for Brainstorming Only:** You may use AI tools on your assignments for brainstorming only. This means that you can use AI to generate some initial ideas or inspiration. However, you must not copy or submit any content produced by AI as your own work. You must clearly indicate in your citations when and how you used AI for brainstorming purposes. It is an honor code offense to use AI tools for more than brainstorming without explicit permission.
- **Category 4 - AI for Limited or Specific Tasks:** You may use AI tools as outlined by your instructor in the assignment guidelines. You must clearly indicate in your citations when and how you used AI for these purposes, and you cannot use AI to generate or assist in generating content beyond that outlined in the assignment. It is an honor code offense to use AI tools for more than the strictly outlined tasks without explicit permission.

Accessibility Services

William & Mary accommodates students with disabilities in accordance with federal laws and university policy. Any student who feels s/he may need an accommodation based on the impact of a learning, psychiatric, physical, or chronic health diagnosis should contact Student Accessibility Services (SAS) staff at 757-221-2509 or at sas@wm.edu to determine if accommodations are warranted and to obtain an official letter of accommodation. For more information, please visit the [SAS website](#).